MODBUS-RTU Communication Protocol TLH & TFP pressure sensor

1. Overview

The protocol adopts the subset RTU model and RS485 half-duplex working mode from MODBUS, which is complied with MODBUS communication protocol.

2. Data Setting

Port setting: no examination, 8 bits data, 1 stop bit.
For example: 9600, N,8,1 means 9600bps, no examination, 8 digits data, 1 stop bit the port baud rate of this transmitter is 1200, 2400, 4800, 9600, 19200, 38400,
57600, 115200.

CRC Examination polynomial:0xA001.
The data in the data communication process is all processed according to double-byte plastic data. If the data identifies a floating-point number, the decimal point needs to be read to determine the size of the data.

3. Communication Format

1. Example of reading command format (03 function code) A.Send,
reading command format:

Address	Function Code	Data Start (H)	Data Start (L)	Data No. (H)	Data No. (L)	CRC16 (L)	CRC16 (H)
0X01	0X03	0X00	0X00	0X00	0X01	0X84	0X0A

B.Return, reading data format

Address	Function Code	Data Length	Data (H)	Data (L)	CRC16 (L)	CRC16 (H)
0X01	0X03	0X02	0X00	0X01	0X79	0X84

2. Writing command format, like 06 function code A.Send,
reading command format:

Address	Function Code	Data Start (H)	Data Start (L)	Data (H)	Data (L)	CRC16 (L)	CRC16 (H)
0X01	0X06	0X00	0X00	0X00	0X02	0X08	0X0B

B.Return, reading data format

Address	Function Code	Data Start (H)	Data Start (L)	Data (H)	Data (L)	CRC16 (L)	CRC16 (H)
0X01	0X06	0X00	0X00	0X00	0X02	0X08	0X0B

3. Abnormal Reply Return

Address	Function Code	1. Abnormal Code	CRC16 (L)	CRC16 (H)
0X01	0X80+Function Code	0x01(Illicit Function) 0x02(Illicit Data Address) 0x03(Illicit Data)		

4. Supported commands and the meaning of commands and data The MODBUS-RTU protocol command list is as follows:

Example: (read data) 01 03 0000 0001 840A (read address)
Address Function code Start address Number of data CRC Example: (write data) 01 06 0000 0002 080B (write address, 1 is changed to 2) Original address Function code Start address to be changed CRC

Function Code	Data start address	Data Number	Byte	Data Scope	Instruction Meaning
0x03 Function Code Data Reading					
0x03	0x0000	1	2	1-255	Read slave address 010300000001840A
0x03	0x0001	1	2	0-1200 1-2400 2-4800 3-9600 4-19200 5-38400 6-57600 7-115200	Baud Rate Reading 010300010001D5CA
0x03	0x0002	1	2	0-Mpa 1-Kpa 2-Pa 3-Bar 4-Mbar 5-kg/cm2 6-psi 7-mh2o 8-mmh2o	Pressure Unit 01030002000125CA
0x03	0x0003	1		0-0000 1-000.0 2-00.00 3-0.000	The decimal point represents 0-3 decimal points respectively 010300030001740A
0x03	0x0004	1	2	-32768-32767	Measurement output value 010300040001C5CB
0X03	0x0005	1	2	-32768-32767	Transmitter range zero point 010300050001940B
0x03	0x0006	1	2	-32768-32767	Transmitter range full point

					010300060001640B
0x03	0x000c	1	2	-32768-32767	Zero offset value, the factory is generally 0 0103000C00014409
0x06 Function code data					
0x06	0x0000		2	1-255	Change the address 010600000002080B
0x06	0x0001		2	0-1200 1-2400 2-4800 3-9600 4-19200 5-38400 6-57600 7-115200	Change baud rate 010600010000D80A 01060001000119CA 01060001000259CB 010600010003980B 010600010004D9C9 0106000100051809 0106000100065808 01060001000799C8
0x06	0x000c		2	-32768-32767	Zero offset. Pressure output =Checked measuring data + Zero offset
Save and reset					
0x06	0x000f		2	0-Save to user area	0106000F0000B9C9
0X06	0x0010		2	1-factory reset	01060010000149CF

Description:

1. When the baud rate is modified, the transmitter will reply to the modified data with the baud rate sent by the host. After the reply, the baud rate of the transmitter will become the modified target value.
2. When modifying the address, the data is also returned to the address before the modification, and the transmitter address will be automatically modified after the reply.
3. Save and reply the factory command will return the original value, indicating that the transmitter has accepted the command from the host.
4. When restoring the factory data, pay attention to the fact that the parameters saved by the factory may be inconsistent with those saved by the user, so the address, baud rate, and calibration data may be inconsistent, so you must search for the transmitter again after restoring the factory parameters.
5. There are only 3 data that the user allows to modify, which are address, address, baud rate, and zero offset value.
6. General users are not allowed to modify the calibration data of the transmitter. If you need to calibrate and modify, please contact our company to obtain the

transmitter calibration software. Sending the command to modify the calibration data by the user will cause the transmitter to output an abnormal code. If you need to modify the calibration data, please use our company's calibration software.

7. If the data that needs to be read is identified by a floating-point number, such as 6.000. However, this protocol stipulates that the data is communicated with plastic data, so the data read is 6000, and then the calculation should be performed according to the position of the decimal point, You can get 6. 000.For example, if the decimal point is 3, it means 6000/10(3), which is 6000 divided by 10 to the third power, and you get the data of 6.000.

8. For detailed verification, please refer to the customer evaluation software provided by our company.